

TARGET-90

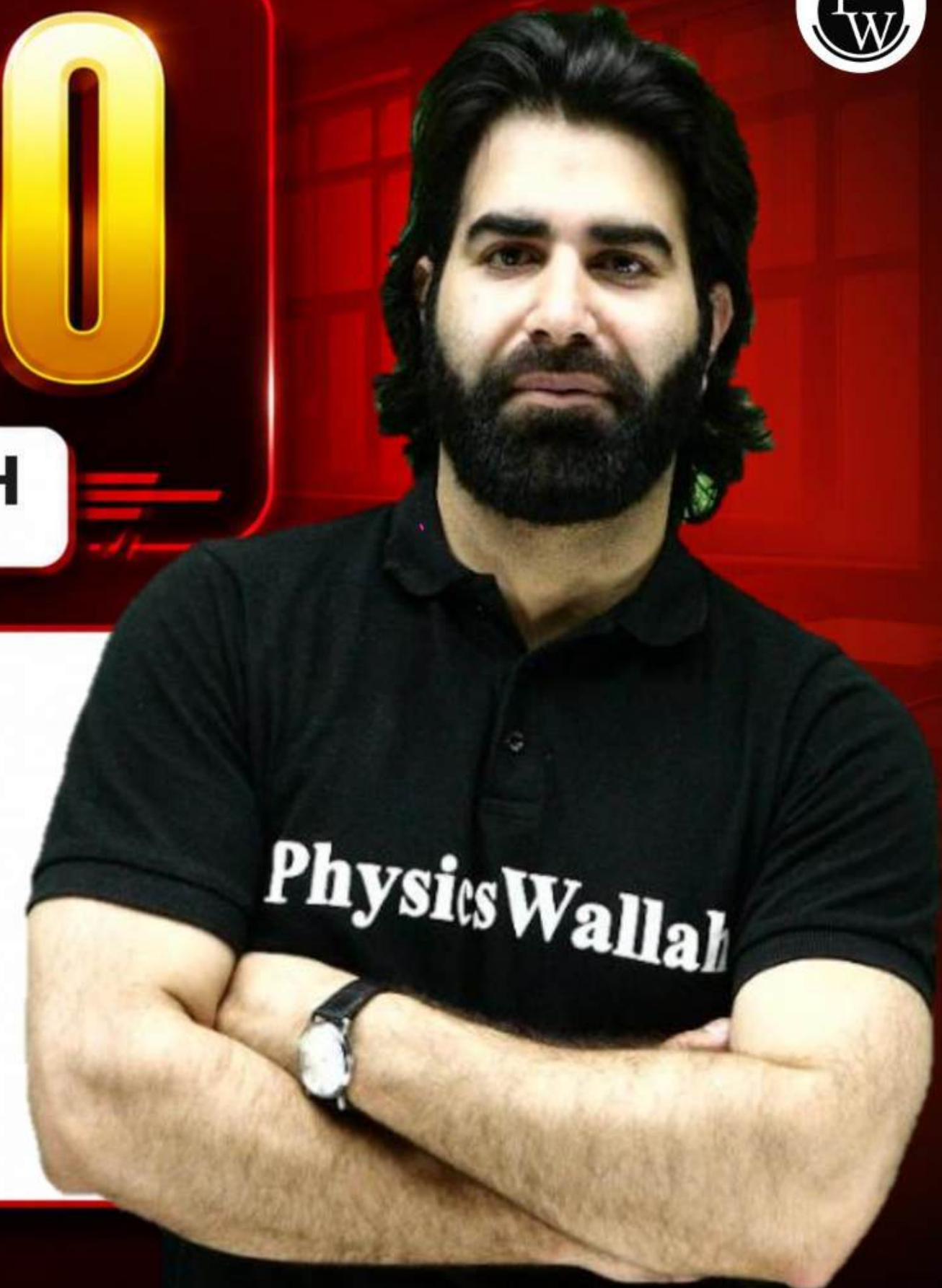
JKBOSE CLASS-11TH

Cell Cycle and
cell Division

One Shot

zoology

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Topics To Be Covered

- 1 Introduction
- 2 Cell Cycle and Division
- 3 Mitosis
- 4 Meiosis.

Summary

cell cycle

R. Virchow

↓
- cell is from
Existing
cell

Birth (formation)



growth



Can Divide

Two ways

→ Mitosis -

→ Meiosis

↓ Some

Remains in
Undivided Stage.



→ New cycle



Cell.

Nucleus

DNA

(Non-Dividing)

(Resting)

cell Division

(Chromatin)

chromosomes





CELL CYCLE

46 chromosome

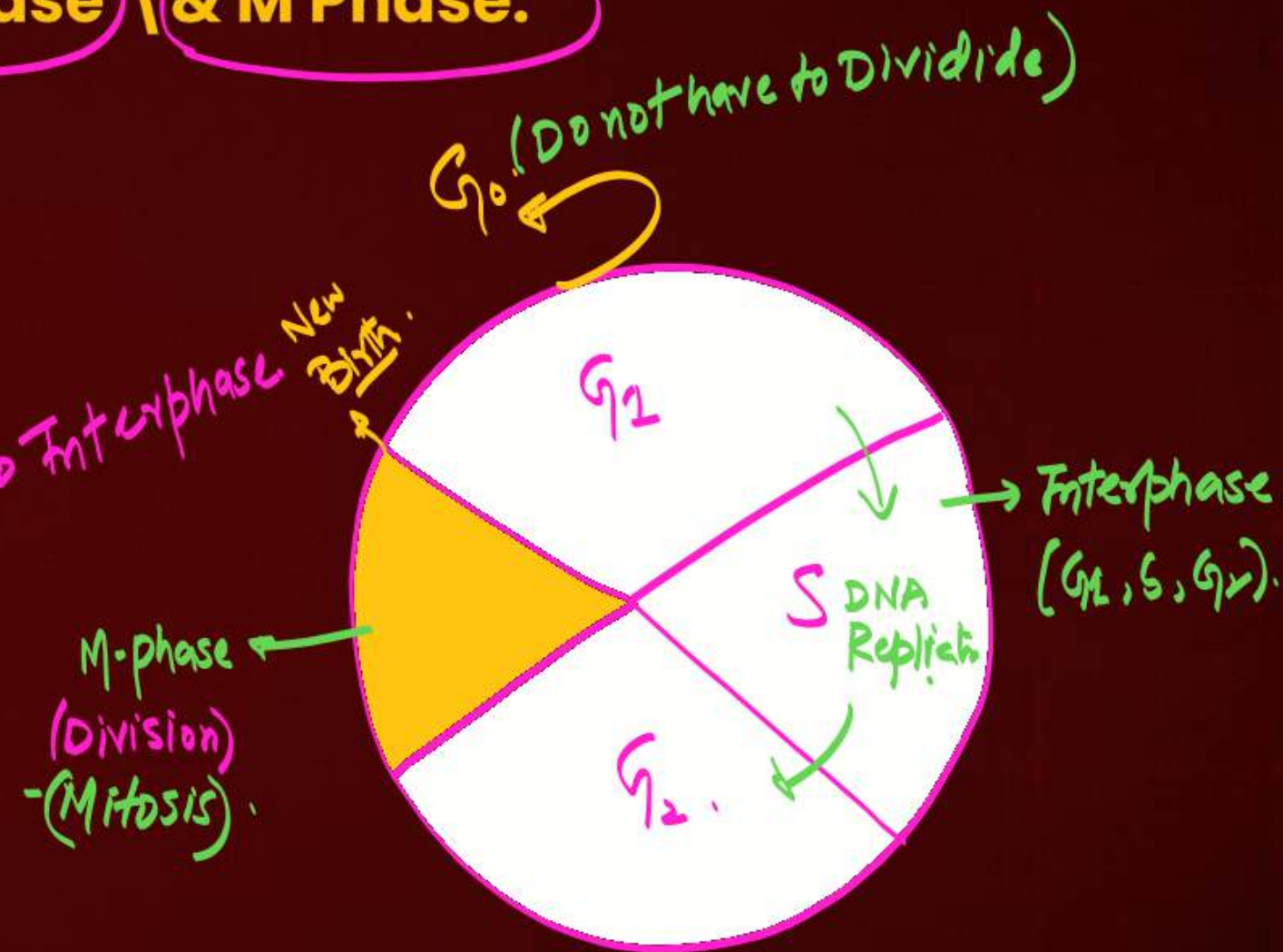
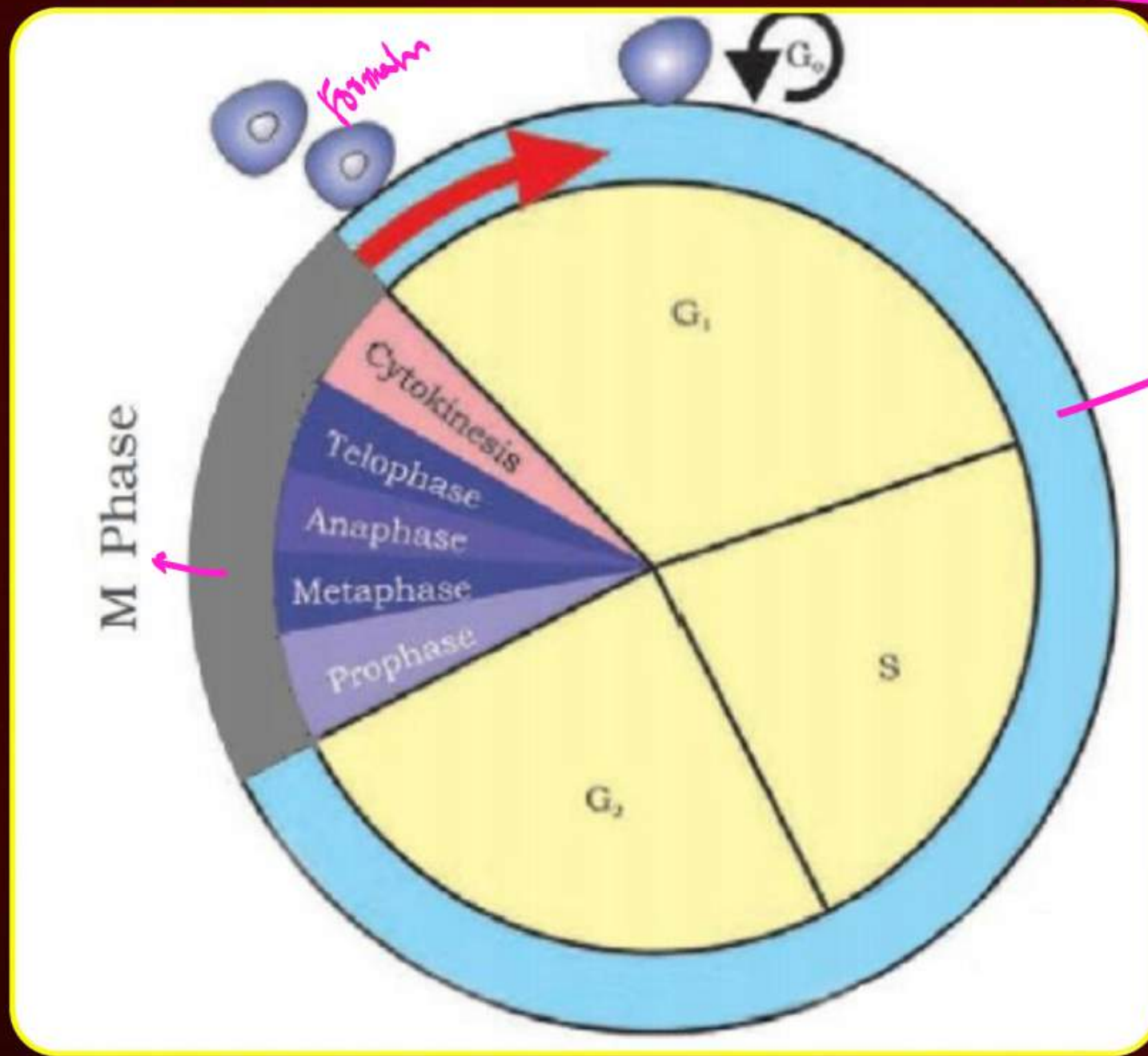


- ❑ It is the life period of a cell during which a cell synthesizes DNA (replication), grows \ divides into 2 daughter cells.
- ❑ Cell growth (cytoplasmic increase) is a continuous process but DNA synthesis occurs only at a specific stage.
- ❑ Duration of cell cycle varies in each organism and each cell type. E.g. Duration of a typical eukaryotic cell cycle (e.g. human cell) is about 24 hrs . In Yeasts, it is 90 minutes.



Phases of Cell Cycle

Cell cycle includes 2 basic phases: Interphase & M Phase.





1. Interphase (resting phase)

- ❑ It is the phase between two successive M phases.
- ❑ It includes cell growth and DNA synthesis.
- ❑ It lasts more than 95% of the duration of cell cycle.

Interphase has 3 phases: $\rightarrow G_1, S, G_2$.



1. Interphase (resting phase)

a. **G1phase (Gap 1 or Antephase):** First growth phase. It is the interval between mitosis and DNA replication. *Mphase & Sphase*

- ☐ Main events:
- ☒ Continuous growth of cell.
- ☒ Cell becomes metabolically active.
- ☒ Prepares machinery for the DNA replication.
- ☒ Synthesizes RNA and proteins.

DNA
↓
RNA
↓
Protein

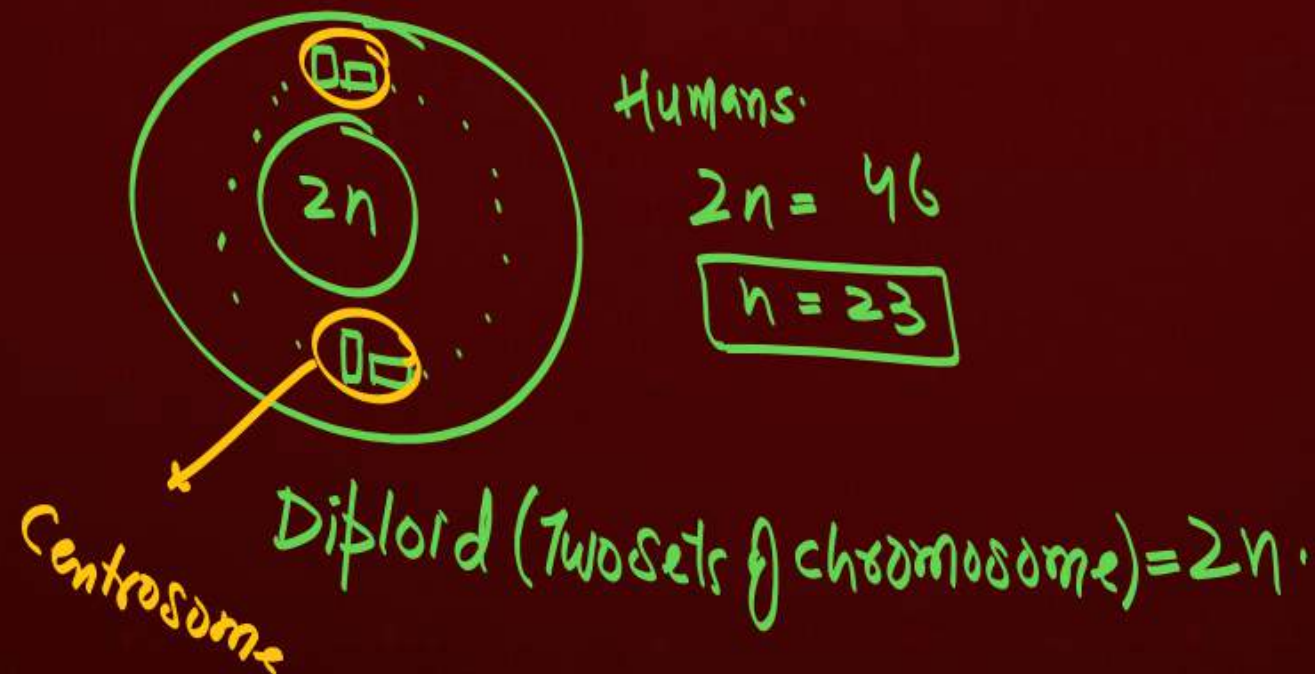
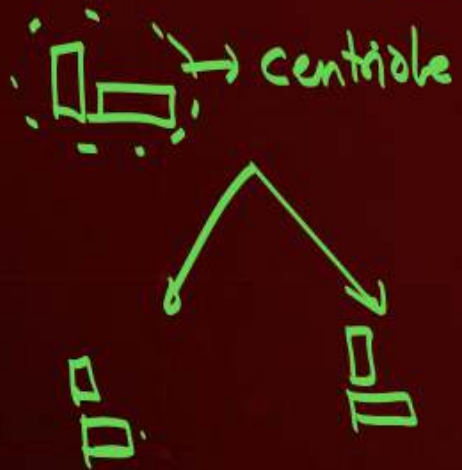


1. Interphase (resting phase)

b. S (Synthetic) phase:

- ❑ In this, DNA replication takes place.
- ❑ Amount of DNA per cell doubles. But chromosome number is not increased.
- ❑ In animal cells, replication begins in the nucleus, and the centriole duplicates in the cytoplasm.

$\frac{46 \text{ chromosome}}{1(\text{DNA})} \xrightarrow{\text{DNA Replication}} \frac{46 \text{ chromosomes}}{2(\text{DNA})}$



Haploid (n) = 23.



1. Interphase (resting phase)

c. G2phase (Gap 2):

- ❑ Second growth phase. Cell growth continues.
- ❑ Synthesis of RNA and proteins continues.
- ❑ Cell is prepared for mitosis.

[1 pocket]
10 Rupee

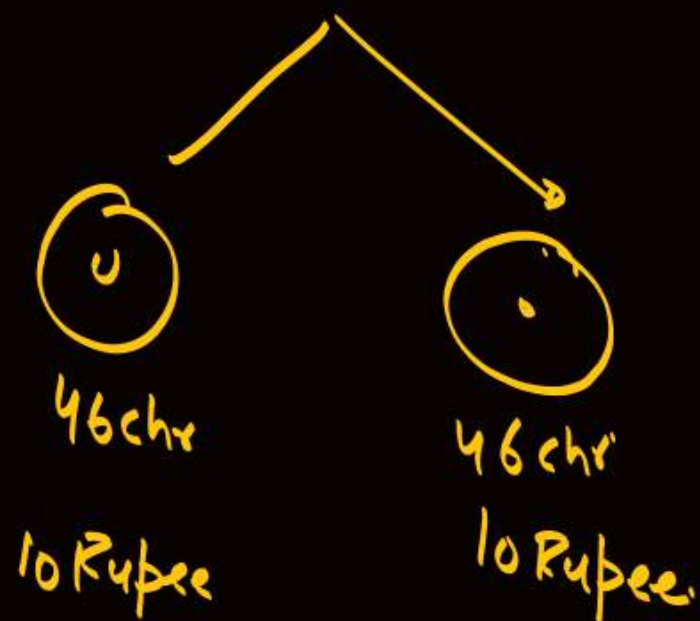


[1 pocket]
20 Rupees

Cell Division is of two types

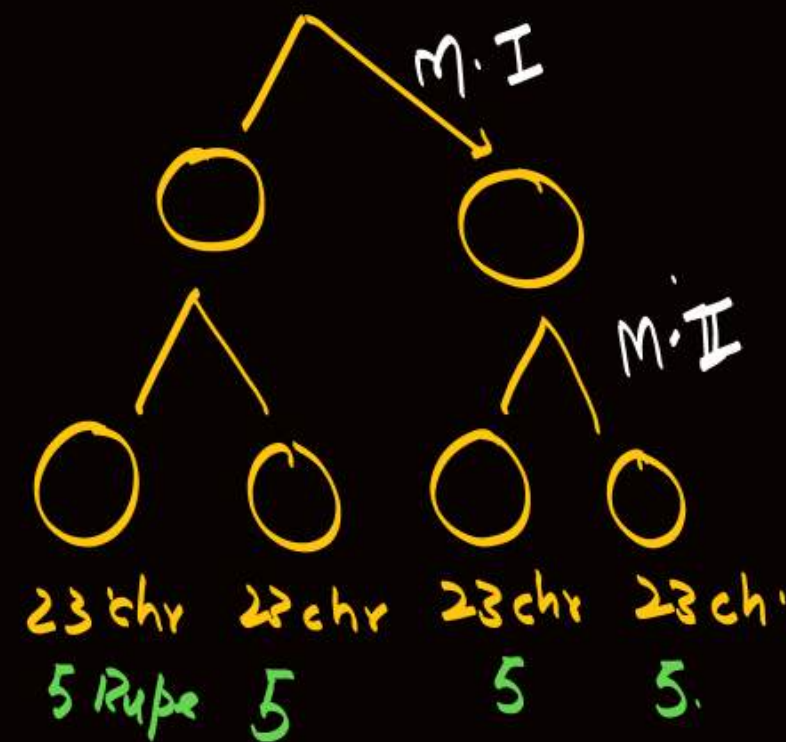
Mitosis

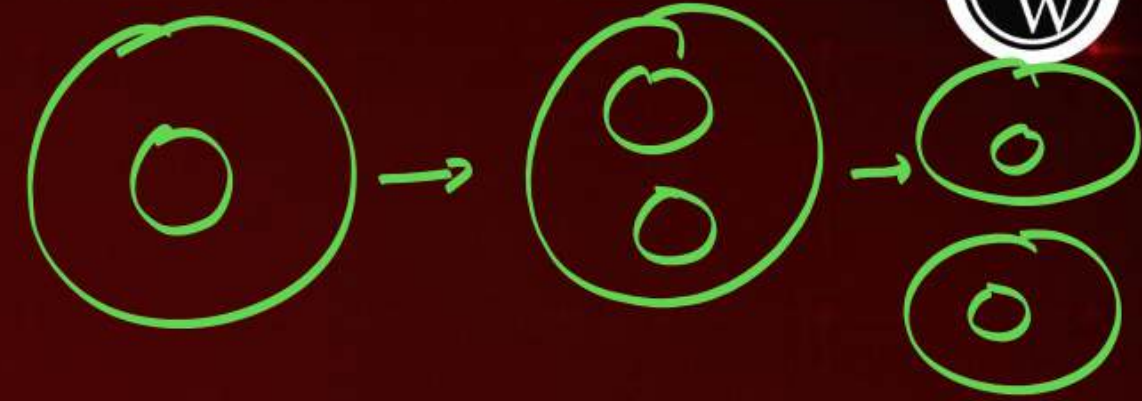
→ Equal No. of chr in Daughter cell



Meiosis [Gamete Forming cell = germ cell]

→ Half No. of chr





2. M Phase (Mitosis phase)

- ❑ It represents the actual cell division (mitosis).
- ❑ In human cell cycle, it lasts for only about an hour.
- ❑ M Phase includes karyokinesis (nuclear division) and cytokinesis (division of cytoplasm).
- ❑ Some cells do not show division. E.g. heart cells. *Skeletal muscle, Cardiac muscle.*
- ❑ Many other cells divide only occasionally to replace damaged or dead cells.
- ❑ The cells that do not divide further exit G_1 phase and enter an inactive stage called quiescent stage (G_0). Such cells remain metabolically active but do not proliferate. *(Divide).*



MITOSIS



Body cells
Somatic cell. → germ cells

- ❑ It is the cell division occurring in somatic cells.
- ❑ It is also called as equational division as the number of chromosomes in the parent and progeny cells is same.
- ❑ Mitosis is generally seen in diploid cells. It also occurs in haploid cells of some lower plants and some social insects. $(2n) = (2n)$
 $(n) = (n)$
- ❑ It involves major reorganization of all cell components.
- ❑ The karyokinesis of mitosis has 4 stages: Prophase, Metaphase, Anaphase & Telophase.



MITOSIS

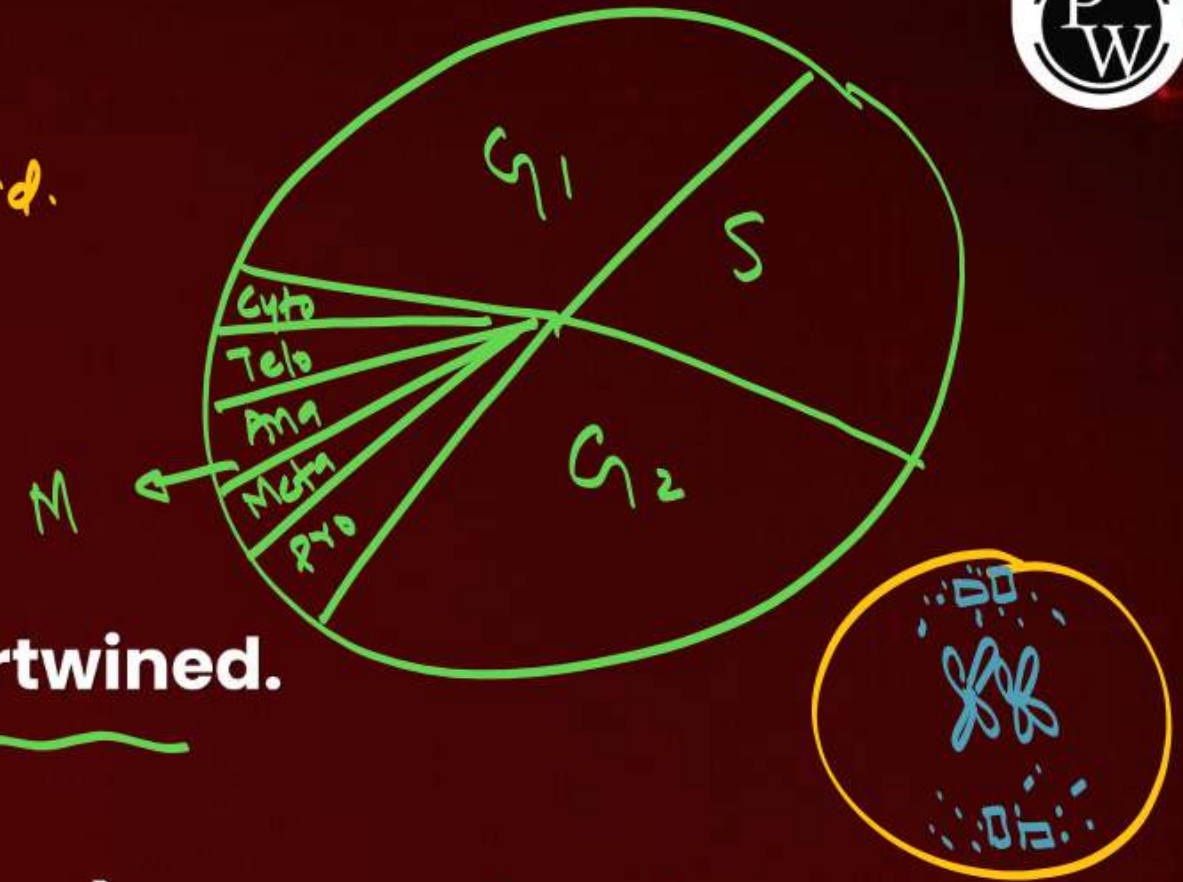
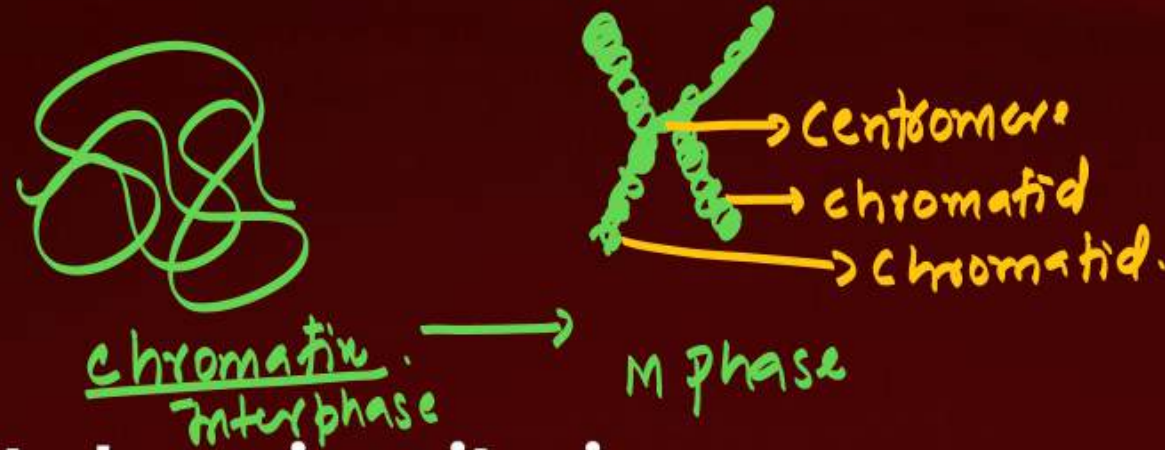


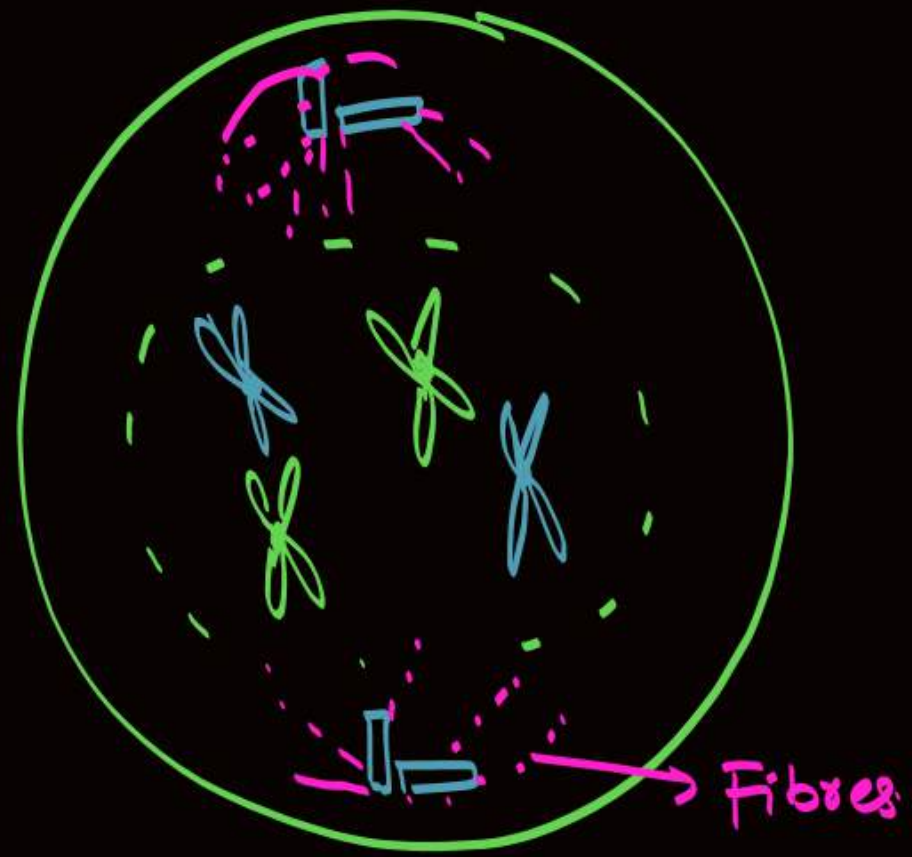
1. Prophase

- It is the longest phase in mitosis.
- It follows the S and G₂ phases of interphase.
- In the S & G₂ phases, DNA molecules are intertwined.

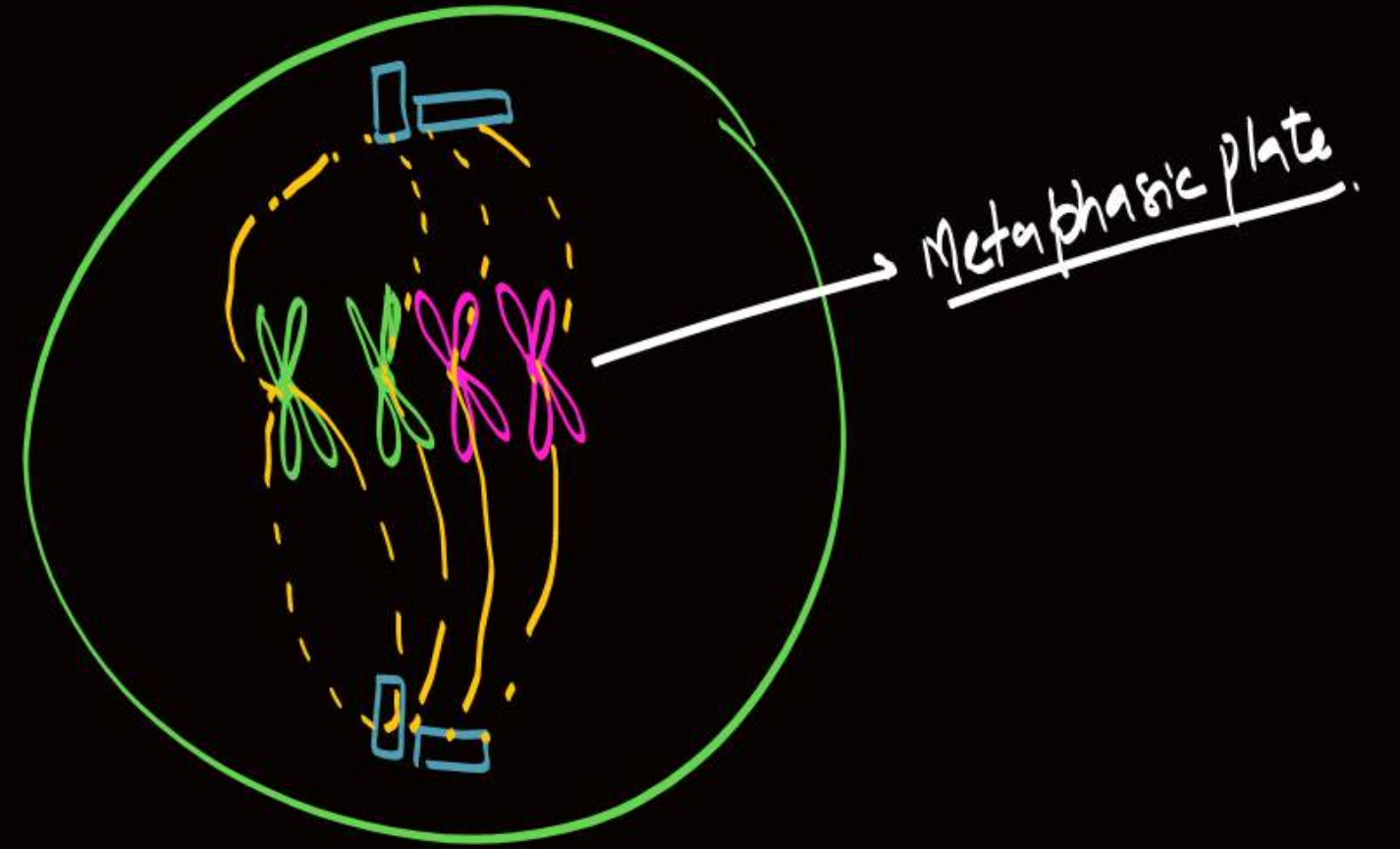
Characteristic events: ✓

- Chromosomal materials (chromatin fibres) are untangled and condensed to form mitotic chromosomes. They are seen to be composed of two chromatids attached together at the centromere.
- Centrosomes begin to move towards opposite poles of the cell. Each centrosome radiates out microtubules called asters. The two asters together with spindle fibres forms mitotic apparatus.





Prophase.





MITOSIS



- ❑ Cells at the end of prophase do not show Golgi complexes, endoplasmic reticulum, nucleolus & nuclear envelope.

2. Metaphase

- ❑ The nuclear envelope completely disintegrates. Hence the chromosomes spread through the cytoplasm of the cell.
- ❑ Chromosome condensation is completed. They can be observed and studied easily under the microscope. They will have two sister chromatids.
- ❑ Chromosomes come to lie at the equator. The plane of alignment of the chromosomes at metaphase is called the metaphase plate.
- ❑ The spindle fibres from both poles are connected to chromatids by their kinetochores in the centromere.

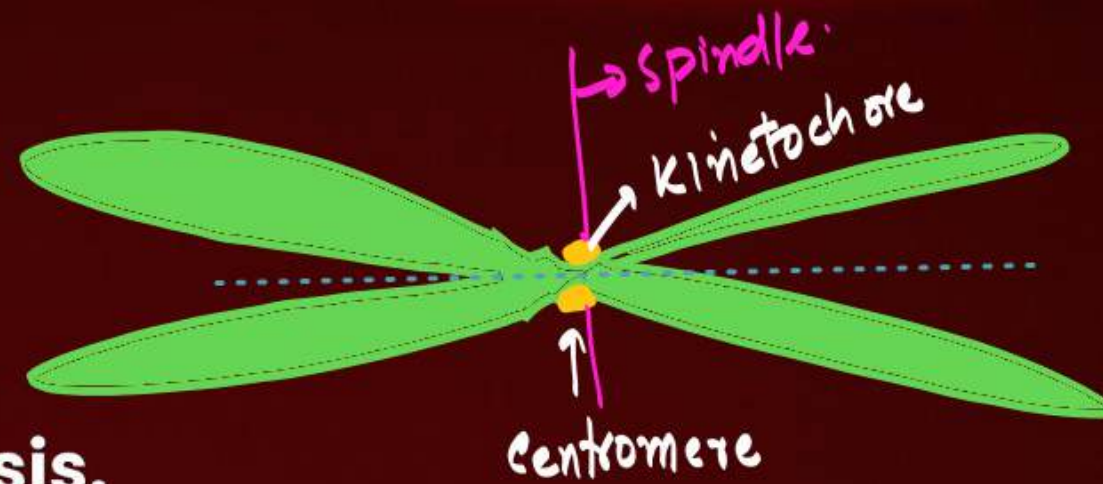


MITOSIS



3. Anaphase

- ❑ It is the shortest phase in the mitosis.
- ❑ Centromere of each chromosome divides longitudinally resulting in the formation of two daughter chromatids (chromosomes of the future daughter nuclei).
- ❑ As the spindle fibres contract, the chromatids move from the equator to the opposite poles.





MITOSIS



4. Telophase

→ opposite of Prophase

- ❑ Chromosomes cluster at opposite poles and uncoil into chromatin fibres.
- ❑ Nuclear envelope develops around the chromosome clusters at each pole. Thus 2 daughter nuclei are formed.
- ❑ Nucleolus, Golgi complex and ER reappear.
- ❑ The spindle fibres disappear.

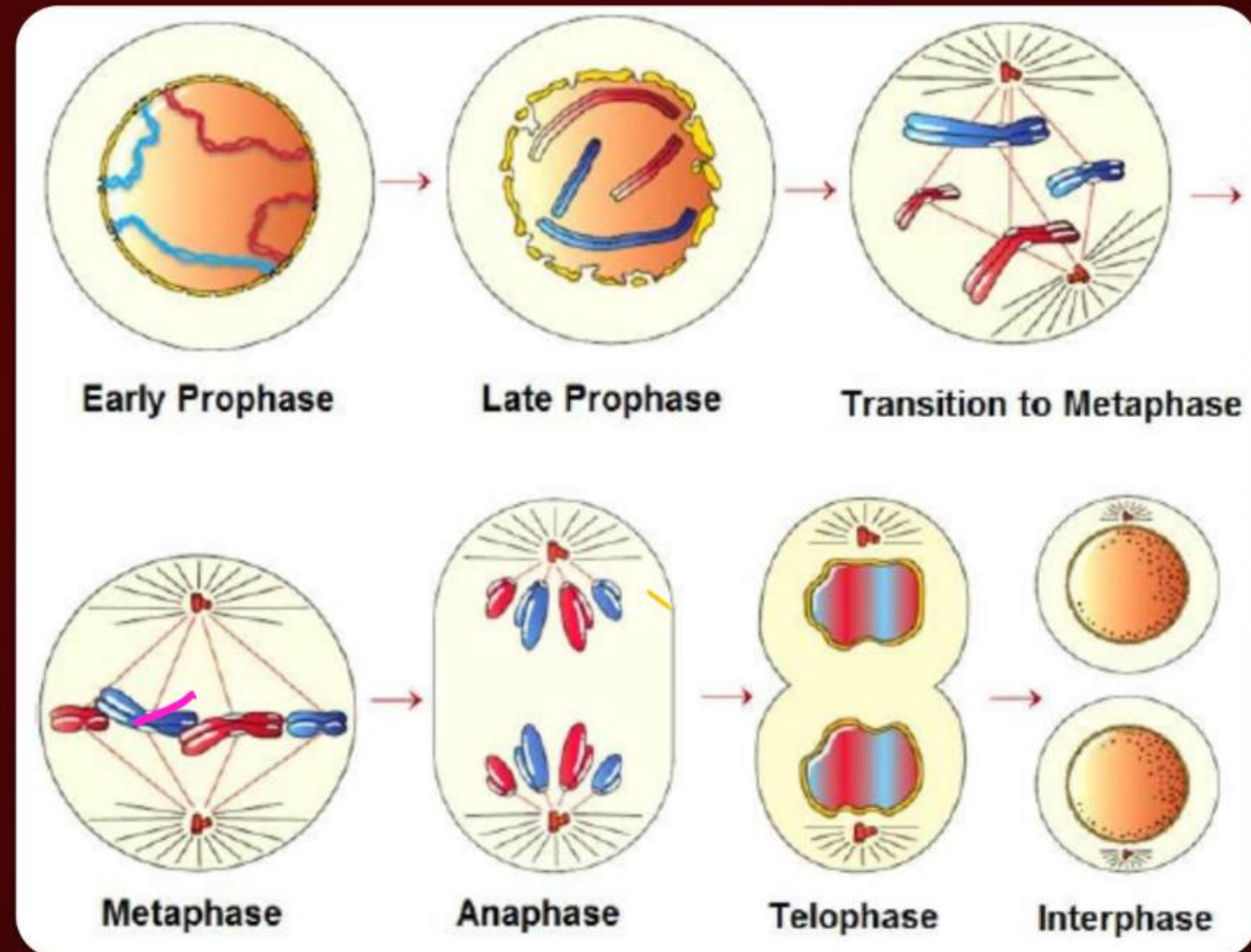


MITOSIS



Cytokinesis

- It is the division of cytoplasm to form 2 daughter cells. It starts when telophase is in progress.



Q no 1 → chromosomes can be Best Studied in which phase.

- (a) Prophase
- (b) Metaphase (✓) → Completely Condensed.
- (c) Telophase
- (d) Anaphase.

Q no 3 - Nuclear Membrane, ER, nucleolus disappears in

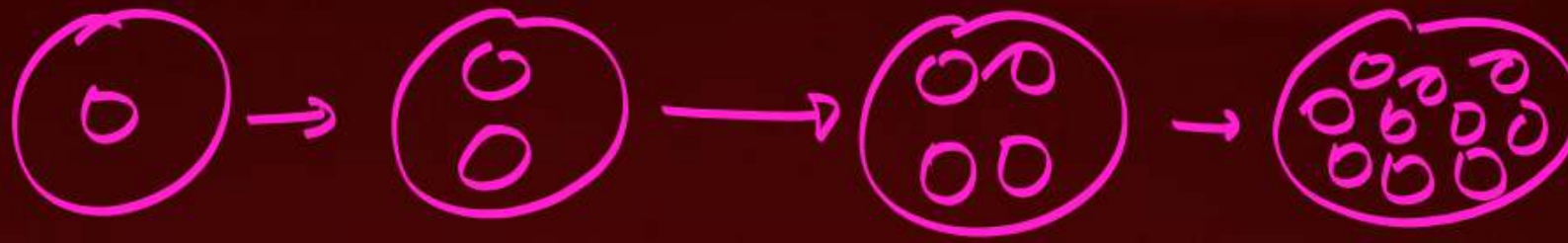
- (a) metaphase
- (b) Anaphase
- (c) Telophase
- (d) Prophase ✓

Q no 2 chromosomes / chromatids Start Moving From Equator to Poles in

- (a) Anaphase (✓)
- (b) Telophase - (Reached)
- (c) Prophase
- (d) Metaphase.



MITOSIS



- During cytokinesis, organelles like mitochondria and plastids get distributed between the daughter cells.
- In some organisms karyokinesis is not followed by cytokinesis. As a result, multinucleate condition (syncytium) arises. E.g. liquid endosperm in coconut.

Significance of Mitosis

- It produces diploid daughter cells with identical genome.
- It helps to retain the same chromosome number in all somatic cells.
- It helps in the body growth of multicellular organisms. Mitosis in the meristematic tissues helps in a continuous growth of plants throughout the life.

Skeletal muscle.

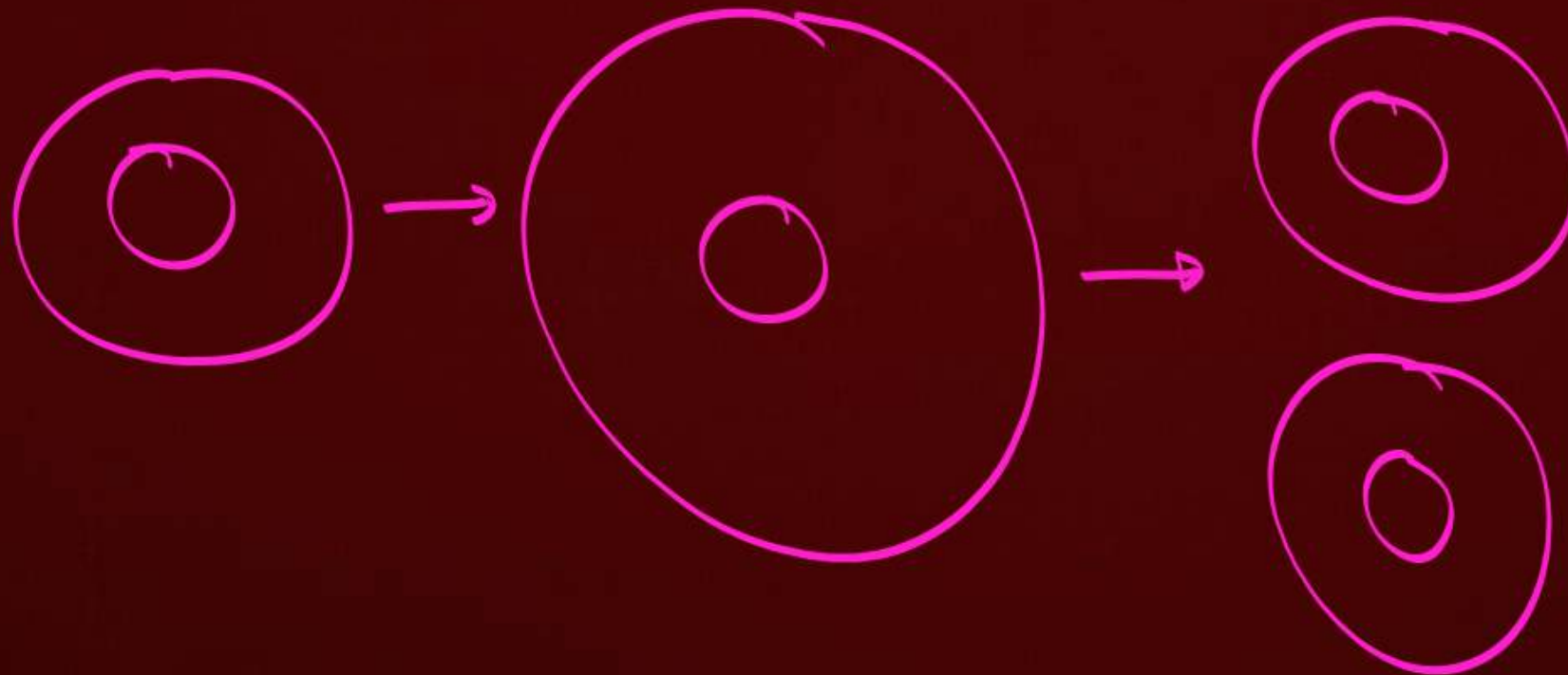
Zygote → Adult
100 Trillions



MITOSIS



- ❑ It restores the nucleo-cytoplasmic ratio that disturbed due to cell growth.
- ❑ It helps in cell repair & replacement. E.g. cells of the upper layer of the epidermis, lining of the gut & blood cells.





Meiosis



Key features of meiosis

Last

- ❑ It involves two cycles (meiosis I & meiosis II) but only a single cycle of DNA replication.
- ❑ It involves pairing of homologous chromosomes and recombination between their non-sister chromatids.
- ❑ Meiosis I begins after replication of parental chromosomes to form identical sister chromatids at the S phase.
- ❑ 4 haploid cells are formed at the end of meiosis II.

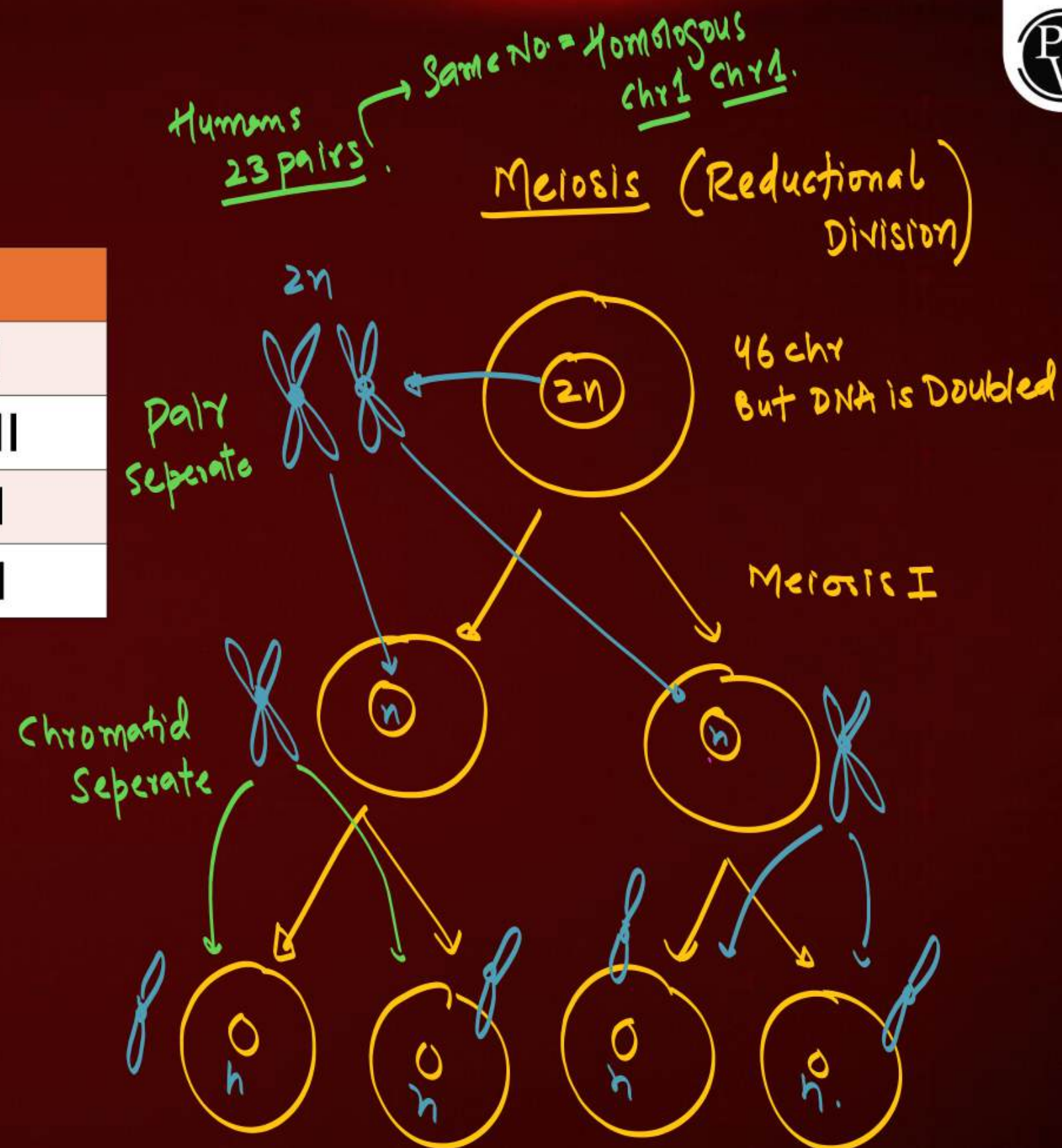


Meiosis

Meiosis I	Meiosis II
- Prophase I	- Prophase II
- Metaphase I	- Metaphase II
- Anaphase I	- Anaphase II
- Telophase I	- Telophase II

Cytokinesis I

Cytokinesis



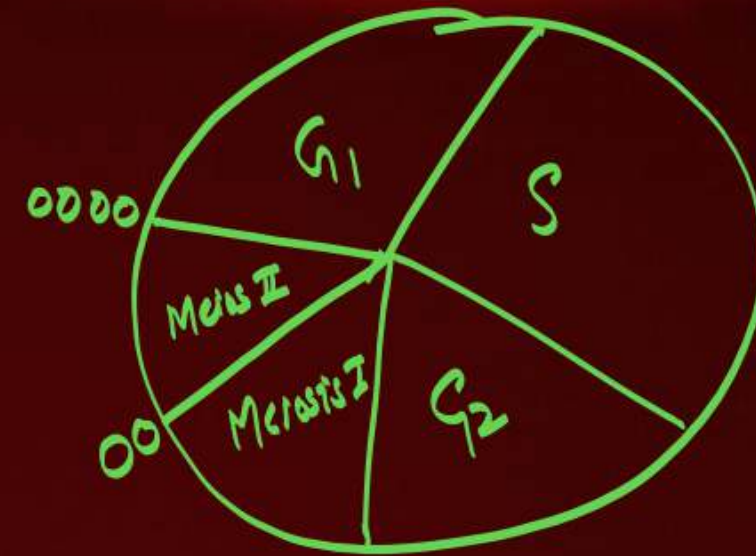


Melosis I



Prophase I:

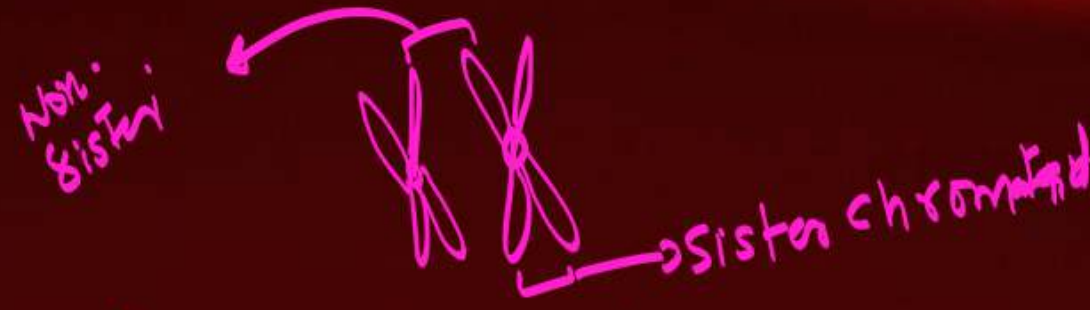
- It is typically longer and more complex.
- It includes 5 phases based on chromosomal behaviour: Leptotene, Zygotene, Pachytene, Diplotene & Diakinesis.
- Leptotene (Leptonema): Chromatin fibres become long slender chromosomes. Nucleus enlarges.
- Zygotene (Zygonema): Chromosomes become more condensed. Similar chromosomes start pairing together (synapsis) with the help of a complex structure called synaptonemal complex. The paired chromosomes are called homologous chromosomes. Each pair of homologous chromosomes is called a bivalent.



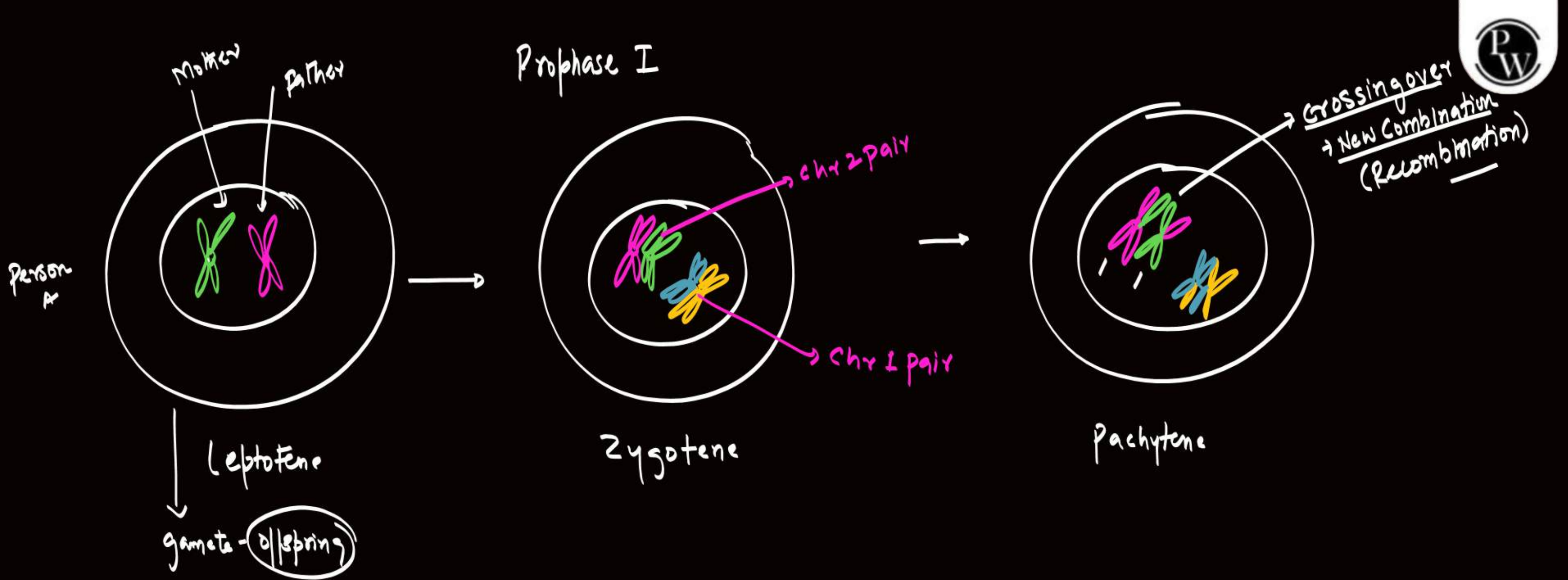
Lib
Liza
Pari
De
Diya



Melosis I



- ❑ **Pachytene (Pachynema):** Comparatively longer phase. Bivalent chromosomes split into similar chromatids. This stage is called tetrads. During this, recombination nodules appear at which crossing over occurs. It leads to genetic recombination on homologous chromosomes.
- ❑ **Crossing over:** The exchange of genetic material between non-sister chromatids of two homologous chromosomes in presence of an enzyme, recombinase.
- ❑ Recombination is completed by the end of pachytene.
- ❑ **Diplotene (Diplonema):** Dissolution of the synaptonemal complex occurs. The recombined homologous



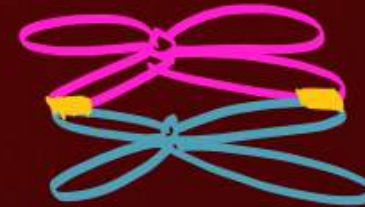


Meiosis I



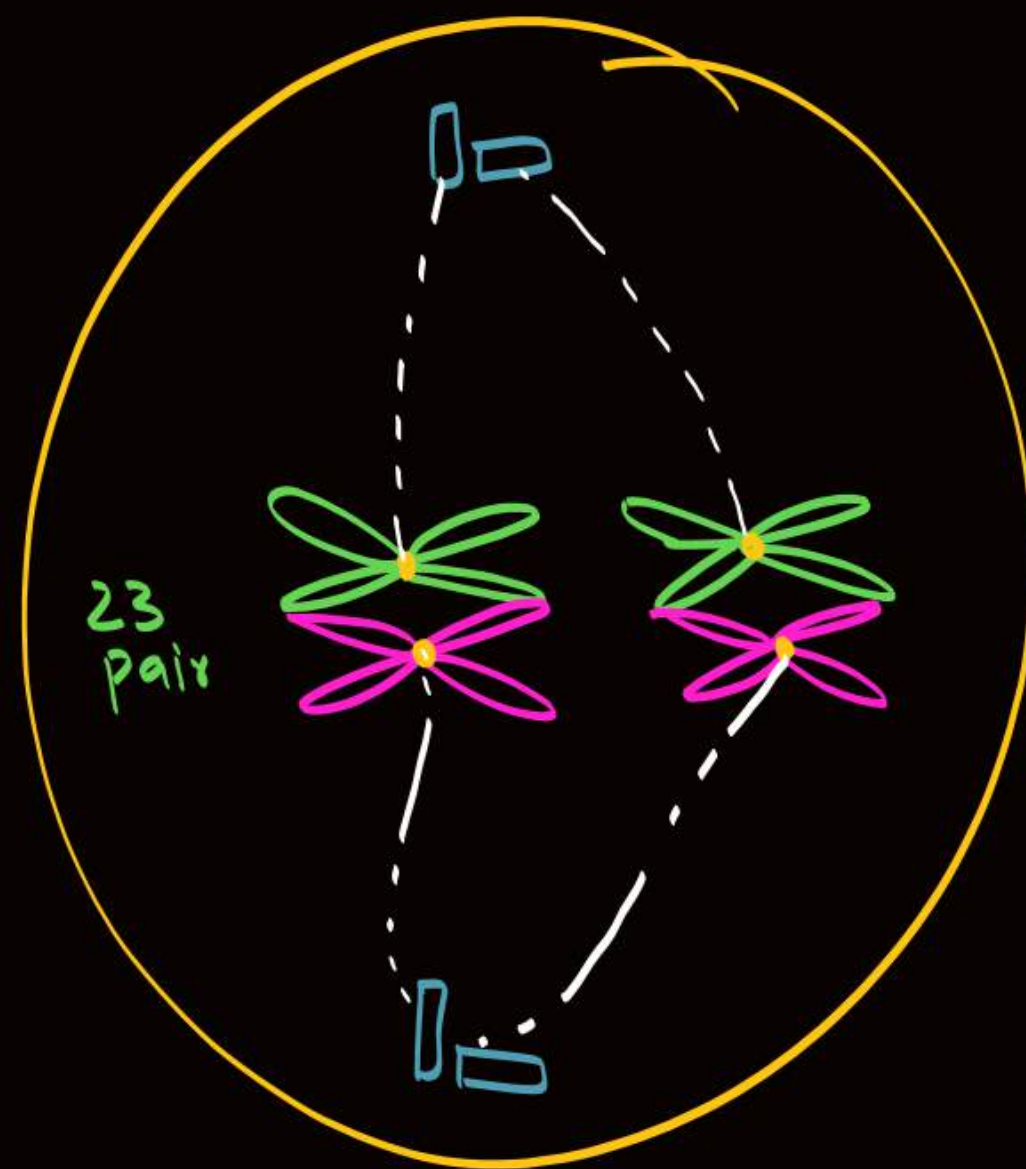
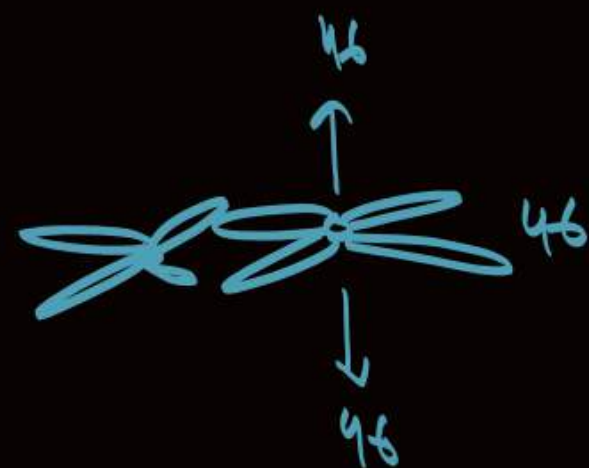
chromosomes of the bivalents separate from each other except at the sites of crossovers. These X-shaped structures are called chiasmata. In oocytes of some vertebrates, diplotene lasts for months or years.

- ❑ Diakinesis: Terminalization of chiasmata. Chromosomes are fully condensed. The meiotic spindle fibers originate from the poles to prepare the homologous chromosomes for separation. Nucleolus & nuclear envelope disappear.

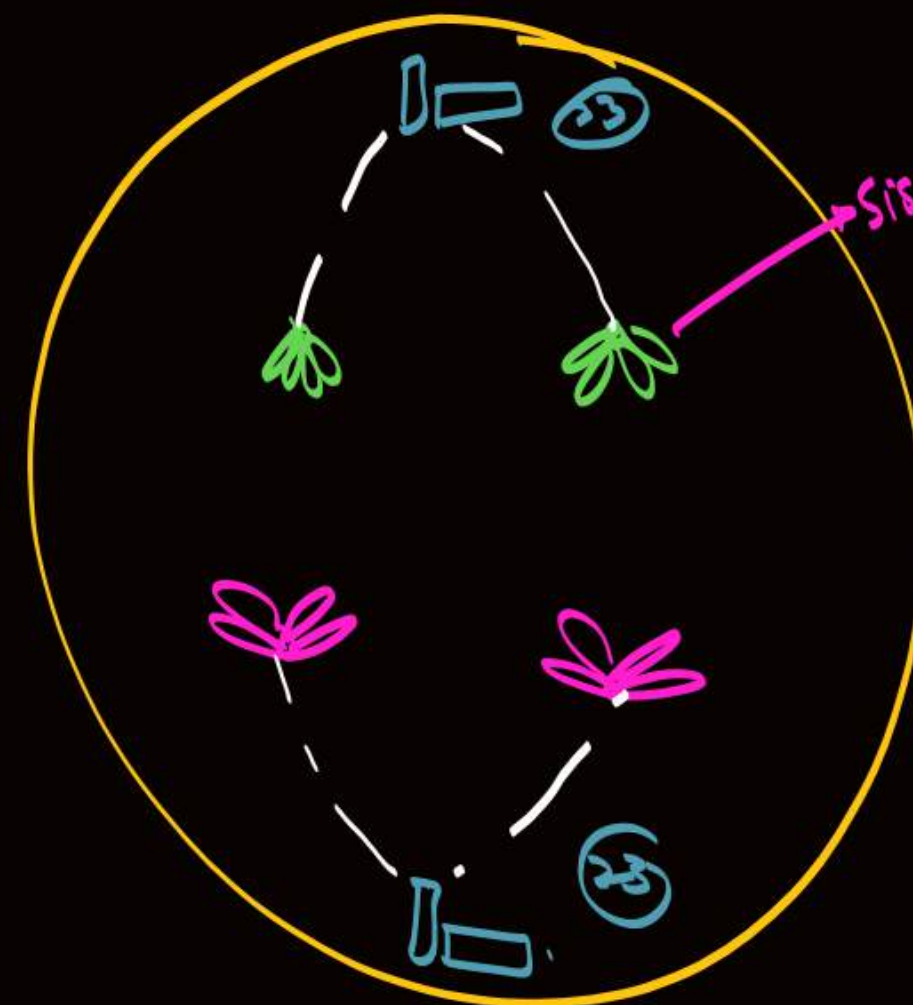


Metaphase I:

- ❑ Spindle formation is completed. The chromosomes align on the equatorial plate. The microtubules from the spindle attach to the pair of homologous chromosomes.



Metaphase I



Anaphase I



Meiosis I



Anaphase I:

- ❑ The homologous chromosomes separate, while sister chromatids remain associated at their centromeres.

Telophase I:

- ❑ The nuclear membrane and nucleolus reappear and 2 haploid daughter nuclei are formed. This is called diad.
- ❑ After this, cytokinesis may or may not occur.
- ❑ After a short interphase, it is followed by meiosis II.
- ❑ This short stage between the two meiotic divisions is called interkinesis.
DNA replication does not occur in this phase.



Meiosis II



- ❑ It resembles the mitosis. It has the following phases:

Prophase II:

- ❑ It is initiated immediately after cytokinesis. The chromosomes again become compact. ✓ ✓
- ❑ Nucleolus and nuclear membrane disappear in both nuclei.

Metaphase II:



- ❑ The chromosomes align at the equator and the microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids.



Meiosis II



Anaphase II:

- ❑ It begins with the simultaneous splitting of the centromere of each chromosome (which was holding sister chromatids together). Thus they move toward opposite poles of the cell by shortening of microtubules attached to kinetochores.

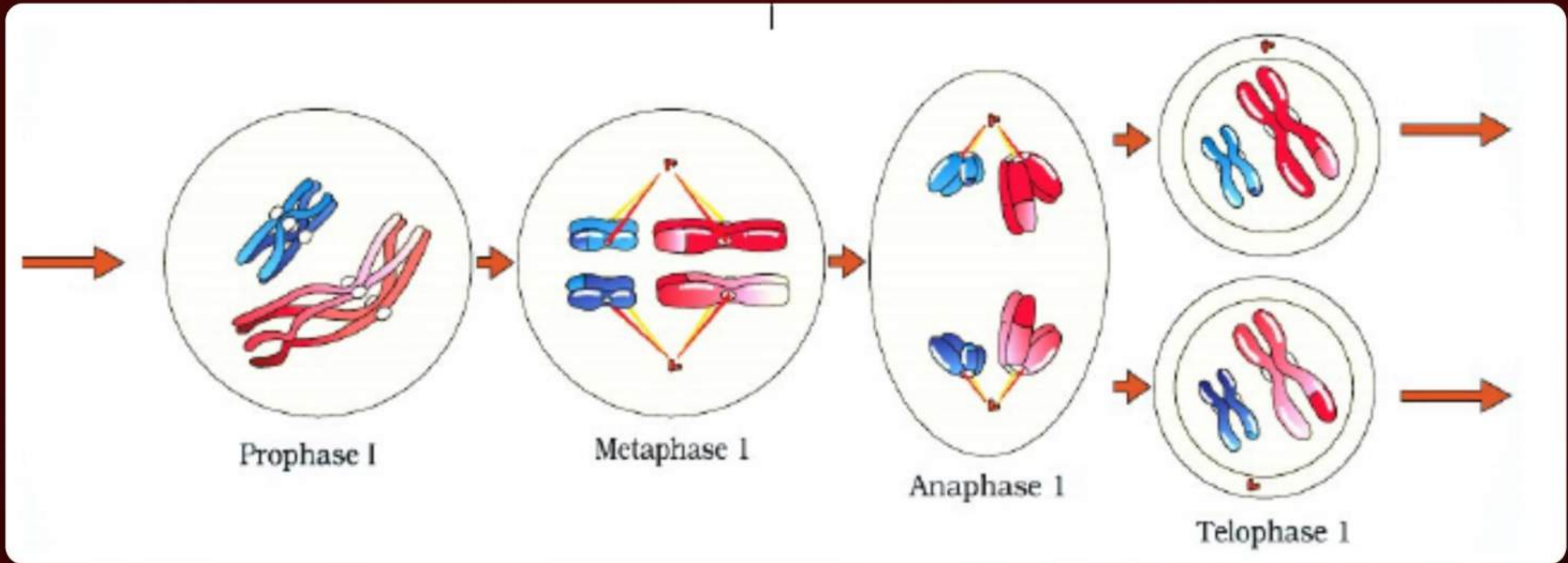
Telophase II:

- ❑ The two groups of chromosomes once again get enclosed by a nuclear envelope; cytokinesis follows resulting in the formation of tetrad of cells i.e., 4 haploid daughter cells.



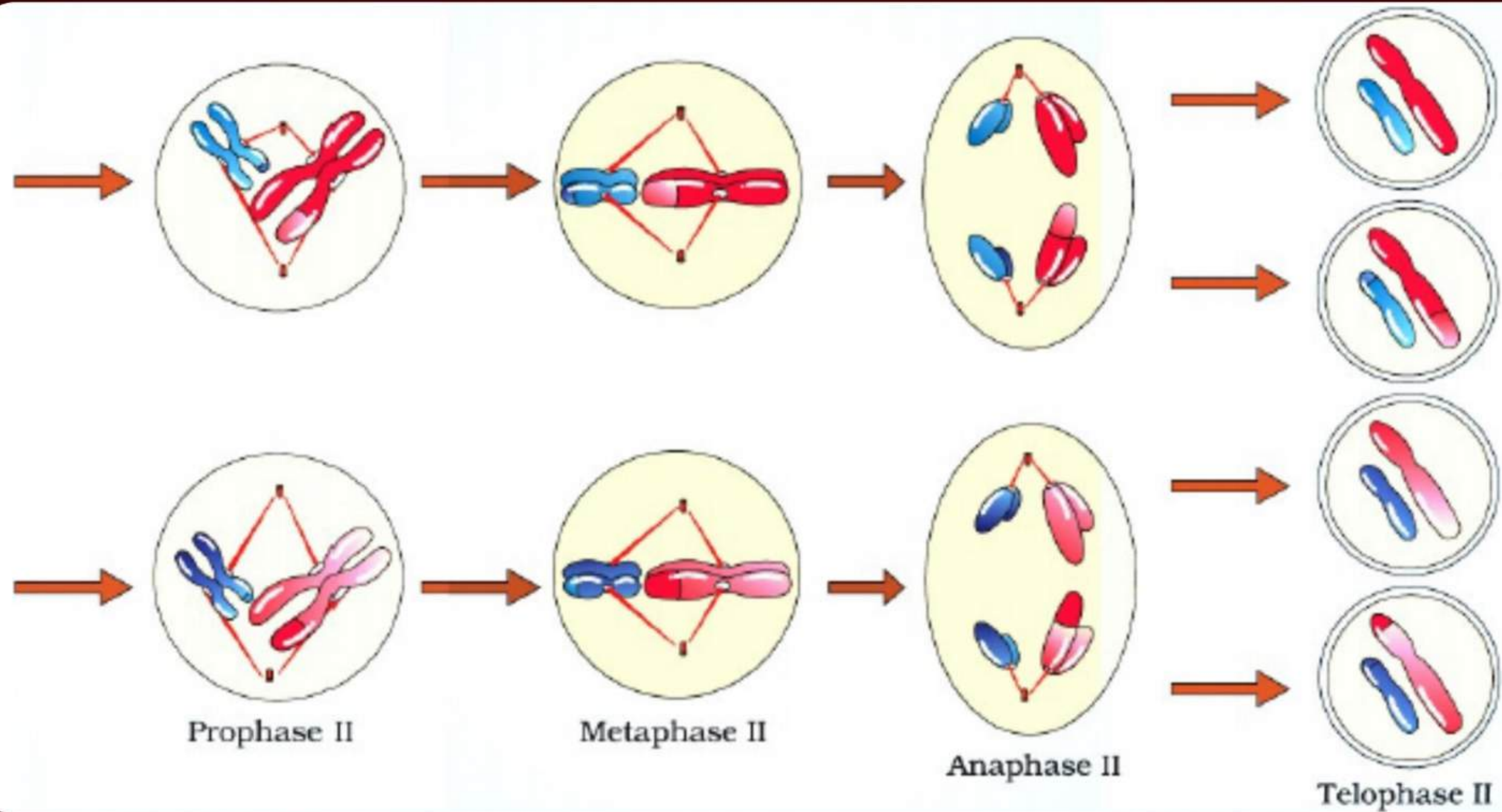
Significance of meiosis

- ❑ It conserves the chromosome number of each species.
- ❑ It causes genetic variation (due to crossing over) in the population of organisms. It is important for evolution.





Significance of meiosis



Writing

Meiosis

Meiosis I, II

Meiosis I - Prophase I

Prophase I - 5 stages

① Leptotene

- Chr condensed

- Nucleus Enlarges

- Chromatid formed.

- pachet

- crossin

En

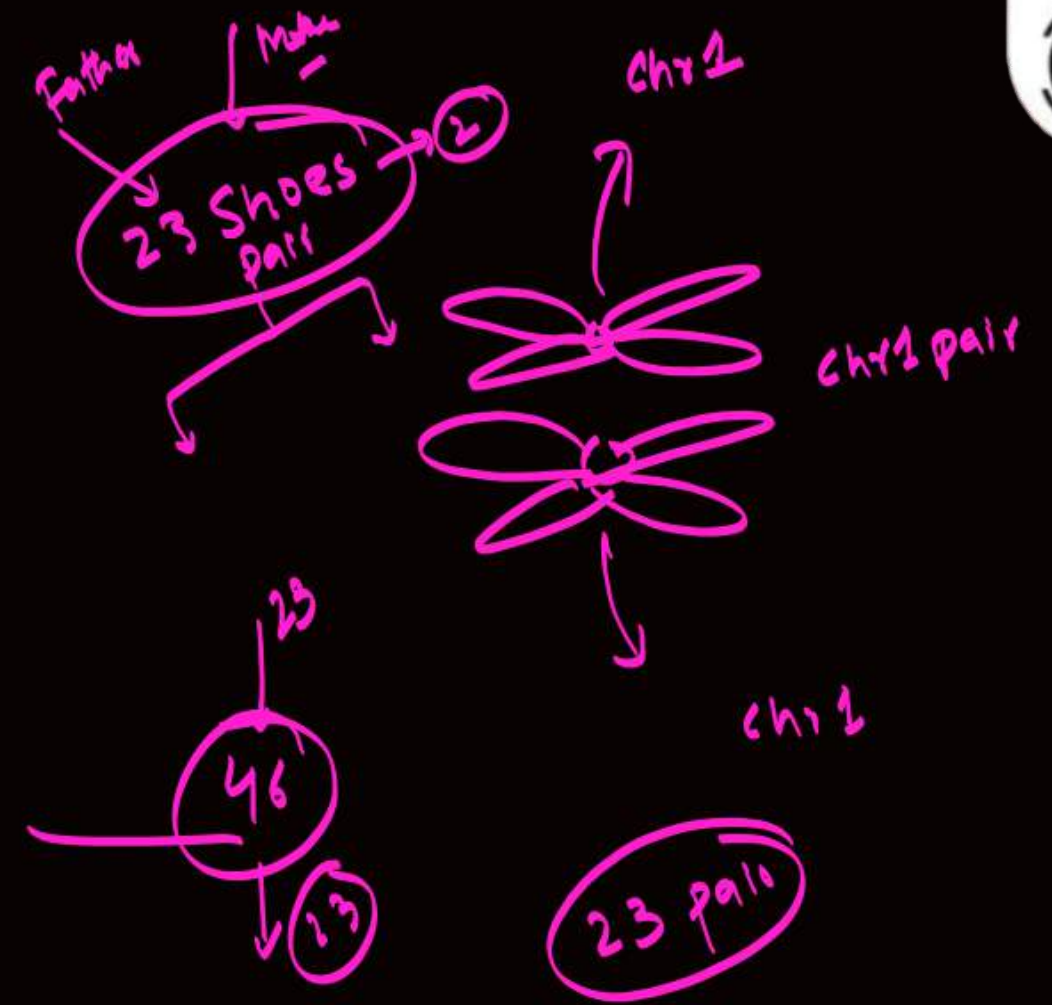
- Diplotene

② Zygotene

- Pairs up

- Synapse

- Syn-cu



Thank
You